The Structural Nature of Strategic Disagreement in Organizations: How the Division of Labor is a Sufficient Explanation for Strategic Disagreement, and How Decision-Makers Can Resolve it

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"I'm not strange, weird, off, nor crazy, my reality is just different from yours" ~Lewis Carroll, Alice's Adventures in Wonderland

1. INTRODUCTION

Organizations accomplish complex tasks by dividing work, assigning employees to specialized roles that concentrate their attention and experience on parts of the organization's operations (Ocasio, 1997). This division of labor is a fundamental structural feature of organizations (Simon, 1947; Joseph & Gaba, 2020), and it is largely efficient: specialization allows employees to develop deep expertise in the functions they oversee and enables organizations to execute at scale. But the division of labor has a less examined consequence, that may ultimately harm organizations' ability to execute strategies.

Particularly, under certain organizational conditions, employees who attend to only a portion of the organizational strategic environment relevant to their role may understand and evaluate their organization's strategy differently. In contexts where strategy is a joint action problem, meaning that collective buy-in is necessary for strategies to be successfully selected and implemented, this difference is likely to be consequential for the organization's performance (Adner & Levinthal, 2024). Because of the potential for the divided attention of employees to hamper strategic success, as well as the clear benefits organizations face when dividing work and thus attention of their employees, it is important to understand when the divided experience of employees leads to disagreement amongst them.

This paper shows that when employees are engaging in causal reasoning about how strategic actions, intermediaries and outcomes are related, an approach recommended by research in the Theory-Based View (TBV) (Felin, Gambardella, & Zenger, 2024; Ehrig & Schmidt, 2022; Singell, 2026), divided attention is itself sufficient to generate fundamental

strategic disagreement among them, even when the proposed strategy employees are evaluating is correct.

The source of this disagreement is what I call the *observational equivalence problem*. When employees observe only a subset of the strategic environment, multiple competing theories about how the strategy works can produce identical predictions for what that employee will see. Because observational data cannot distinguish among possible theories (see Ryall & Sorenson, 2025), the employee's beliefs about causal structure are determined not by evidence, but by the priors formed through their organizational role.

Different employees, observing different parts of the organization, face different versions of this problem — different *equivalence classes* of theories that are indistinguishable from their vantage point. The result is that two employees observing the same causal environment arrive at fundamentally different beliefs about how the strategy works — not because they interpret evidence differently, but because different causal architectures are observationally equivalent from their respective vantage points.

I formalize this argument using Bayesian networks, mathematical representations of theories as causal structures with associated beliefs about the strength of each causal relationship (Pearl, 2009; Camuffo, Gambardella & Pignataro, 2024). A theory in this framework has two components: its *structure* — which variables cause which outcomes — and its *strength* — how much each cause shifts the probability of its effect. Correspondingly, disagreement can arise at two levels.

Within-theory disagreement occurs when employees share the same causal structure but differ on how strong the effects are. This form of disagreement is benign in the sense that all employees agree the proposed strategy is the right experiment to run — they simply disagree

about how large the effect will be. *Between-theory disagreement* occurs when employees hold different beliefs about which variables cause which outcomes — when they disagree about the architecture of the causal world, not merely the magnitudes within it. This is the form of disagreement the paper focuses on, because it cannot be resolved by implementing the proposed strategy and observing the outcome.

The paper's central result is that the division of labor, by restricting employee observation sets in the way described, generates between-theory disagreement as its default outcome whenever the true strategic environment contains causal relationships among variables that different employees attend to. This disagreement is structurally produced: it arises from the same organizational feature that makes specialized execution possible. It does not require bad faith, cognitive bias, or informational asymmetry of the conventional kind. It is a mathematical consequence of the fact that observational equivalence classes differ across observation sets.

I then show how decision-makers can resolve this disagreement. The key insight is that between-theory disagreement requires a qualitatively different type of experiment than within-theory disagreement. Rather than testing whether the proposed action produces the desired outcome — which employees with different structural beliefs will interpret differently — the decision-maker should first test the relationship between the causal antecedents that different employees attend to. This *structurally diagnostic experiment* produces evidence whose outcome is differentially probable under the competing causal architectures, breaking the observational equivalence that prevents employees from updating their structural beliefs. Whether the diagnostic experiment reveals the antecedents to be causally related or independent, it generates a shared causal framework within which the proposed strategy can be fairly evaluated.

The paper makes two principal contributions to research on strategy and cognition. First, it extends the Theory-Based View from a single decision-maker selecting and testing strategies to an organizational setting in which multiple employees with role-constrained observation sets must collectively evaluate and implement those strategies. The extension reveals a coordination cost of theory-based strategy that is invisible in the single-actor framing: the explicitness that makes theories valuable for action selection also makes them grounds for structural disagreement among employees who have been differentially exposed to the causal environment. Second, the paper provides a formal account of *when* and *why* this disagreement arises, and what type of experiment resolves it — moving from a qualitative observation that divided labor generates strategic misalignment to a precise identification of the conditions, the mechanism, and the remedy.

2. MODEL

Adapting work in the theory-based view and causal logic (Camuffo, Gambardella, & Pignataro, 2024; Pearl, 2009), I represent the strategic theories of decision-makers and employees using Bayesian networks, where actors form and update beliefs about both the strength and structure of the relationships between strategic actions and outcomes. I distinguish between the role of the decision-maker, who proposes strategic actions as interventions formalized using the do operator (Pearl, 2009; Ryall & Sorenson, 2025), and the employees, whose limited and situated attention can lead to observations of a strategy that are differentially observationally equivalent (Ocasio, 1997). I then model how these actors evaluate a proposed strategy by calculating the expected value of the desired strategic outcome conditional on their observations and beliefs. Table 1 summarizes the key parameters in the model.

Table 1: Model Parameters

Symbol	Definition
Attributes	
X_S	Strategic action variable available to the decision-maker
X_C	Intermediate attribute that may lie on the causal pathway between X_S and X_T
X_T	Strategic outcome of interest
$\mathcal{O}_i$	Subset of attributes passively observed by employee i, determined by organizational role
Causal Structures	
$\mathcal{G}$	Set of candidate directed acyclic graphs (DAGs) over the relevant attributes
G_k	A single candidate DAG representing one theory of the causal structure; $k = 1, \ldots, n$
$do(X_S = x)$	Pearl's (2000) intervention operator; sets X_S to x by severing all incoming arrows to X_S. When X_S is exogenous, $P(X_T \mid do(X_S = x)) = P(X_T \mid X_S = x)$
Parameters and Parameter Spaces	
θ_k	Vector of conditional probability parameters indexing the strength of causal relationships within G_k
Θ_k	Restricted space of θ_k consistent with the decision-maker's interventional claim under G_k, following CGP (2024)
$\tilde{\Theta}_k$	Null parameter space; θ_k values under which the proposed action has no effect on X_T under G_k
Beliefs	
$\omega_{i,k}$	Employee i's prior probability that G_k is the correct causal structure; $\omega_{i,k} \in (0,1)$, $\sum_k \omega_{i,k} = 1$
$\omega'_{i,k}$	Posterior theory weight after observing $\mathcal{O}_i$; updated via Bayes' rule
$\mu_i(\theta_k \mid G_k)$	Employee i's prior over causal effect strengths within G_k; corresponds to CGP's (2024) $\mu_\Theta(\theta)$
$\mu_i(\theta_k \mid G_k, \mathcal{O}_i)$	Posterior parameter distribution after observing $\mathcal{O}_i$
$\sigma_{\Theta_k}(\theta_k)$	Standard deviation of $\mu_i(\theta_k \mid G_k)$; theory uncertainty following CGP (2024)
Value Functions	
$v_k(\theta_k)$	$P(X_T \mid do(X_S), G_k, \theta_k)$: expected probability of the desired outcome under the proposed intervention given G_k and θ_k
$V_{\Theta_k, i}$	Employee i's expected value of the intervention under G_k, integrating v_k over $\mu_i(\theta_k \mid G_k)$ on Θ_k; corresponds to CGP's (2024) V_Θ
V_i	$\sum_k \omega_{i,k} \cdot V_{\Theta_k,i}$: theory-weight-averaged expected value of the intervention across $\mathcal{G}$

2.1 A Motivating Example

To illustrate how decision-makers and employees form and use theories about strategic actions and outcomes, I use the CEO of Southwest Airlines' proposed change from unassigned to assigned seating, and model the strategic theories formed by the Chief Operating Officer (COO) and the Chief Customer & Brand Officer (CCBO).

Southwest Airlines is a budget airline that relies on a low-cost operating model (Carroll & Sørensen, 2021). One of the major ways that Southwest achieves lower costs than competitors is by operating fewer gates per flight at each airport, which is done by achieving faster turnaround times at the gate. Recently Southwest's turnaround time at the gate has increased,

jeopardizing the low-cost operations of Southwest. Thus, the strategic outcome of interest is the turnaround time at the gate, formalized as a state space $X_T = \{F, L\}$, where F denotes fast turnaround and L denotes slow turnaround.

The CEO's proposed strategic action is to change the airline's seating policy from unassigned to assigned seating. Southwest Airlines historically operated an unassigned seating model intended to contribute to faster boarding and turnaround times. The CEO proposes that switching to assigned seating will restore faster turnaround times. The strategic action seating type is formalized as a state space $X_S = \{U, A\}$, where U denotes unassigned seating and A denotes assigned seating.

Whether changing the seating type will improve turnaround time may depend on other attributes of the strategic environment. The CCBO observes that customer compliance at the gate, whether passengers board in an ordered and timely way, may also impact the speed of turnaround, either directly or as an intermediary between the seating policy and its effect on turnaround time. Customer compliance is formalized as a state space $X_C = \{M, N\}$, where M denotes compliant customers and N denotes non-compliant customers.

The three states spaces $\{X_T, X_S, X_C\}$ constitute the strategic state space for the Southwest motivating example, and are the attributes over which the CEO, COO, and CCBO each form their theories. Generally, the state spaces over which actors form theories of their organization's strategic environment contain some combination of relevant strategic actions (seating type X_S from U→A), strategic outcomes (turnaround at gate X_T) and possible strategic intermediaries (customer compliance X_C).

2.2 Strategic Theories as Bayesian Networks

With the strategic state space established, I formalize how organizational actors form theories, which are representations of their understanding of how strategic actions and intermediaries causally relate to the strategic outcome of interest (Sorenson, 2024). Following Camuffo, Gambardella, and Pignataro (2024) and Pearl (2009), I model strategic theories as Bayesian Networks — directed acyclic graphs (DAGs) over the relevant state space, equipped with conditional probability distributions that quantify the strength of each causal relationship in the graph.

A strategic theory is composed of two objects (G_k, θ_k), that represent the structure and the strength of causal relationships held by an actor. The structure of the theory G_k is a DAG over the state space $\{X_T, X_S, X_C\}$, and represents which variables cause which outcomes. While the strength of these causal relationships is captured by θ_k, and represents how much each cause shifts the probability distribution of it effect. I define each component in turn below before showing how they combine into a complete theory.

2.2.1 The Structure of a Theory

The structure of a theory is formalized through a set of potential theories G, where a candidate theory G_k is a DAG over the state space $\{X_T, X_S, X_C\}$, representing one possible structure of how the actions, intermediaries and outcomes in the strategic environment are causally organized. It specifies which directed causal arrows exist between the strategic variables and in what direction. Given the Southwest state space $\{X_T, X_S, X_C\}$, three candidate DAGs are relevant for theories held in the organization:

$G_1: X_S \rightarrow X_T$ (seating drives turnaround directly)

$G_2: X_S \rightarrow X_C \rightarrow X_T$ (seating drives compliance, which drives turnaround)

$G_3: X_S \rightarrow X_T \leftarrow X_C$ (seating and compliance are independent causes of turnaround)

I denote the full set of candidate graph for Southwest as $G = \{G_1, G_2, G_3\}$. These graphs are not exclusively the CEO's set of potential theories, they represent a range of structurally distinct causal models that any actor, such as the COO and CCBO, in the organization might hold about how the strategic state space is organized. The graph each actor finds the most credible in the set of candidate graphs G is a question of belief, which is formalized in section 2.3 below.

2.2.2 The Strength of a Theory

The strength of a theory is formalized through a parameter vector for each graph θ_k, which indexes the conditional probability distributions $\mu(\theta_k)$ attached to each node in a graph G_k. Each element of θ_k is a conditional probability of the form $P(\text{effect} \mid \text{cause})$ implied by the structure of G_k. In the absence of any theoretical constraint, θ_k can take any value in the open unit interval: each conditional probability parameter ranges freely over (0,1).

For example, in graph G_1 where the seating type causes the turnaround time at the gate directly and where fast turnaround time (F) is the desired outcome, the parameter θ_1 will contain information about the likelihood of fast turnaround time at the gate given unassigned and assigned seating (i.e. $P(F|X_S = A), P(F|X_S = U)$). For the three candidate graphs in the Southwest example, the parameter vectors are:

Under G_1**:** $\theta_1 = (\theta_{F|A}, \theta_{F|U})$, where $\theta_{F|A} = P(F \mid X_S = A)$and $\theta_{F|U} = P(F \mid X_S = U)$

Under G_2**:** $\theta_2 = (\theta_{N|U}, \theta_{N|A}, \theta_{F|M}, \theta_{F|N})$, where $\theta_{N|U} = P(N \mid X_S = U)$, $\theta_{N|A} = P(N \mid X_S = A)$, $\theta_{F|M} = P(F \mid X_C = M)$, and $\theta_{F|N} = P(F \mid X_C = N)$

Under G_3**:** $\theta_3 = (\theta_{F|M,A}, \theta_{F|N,A}, \theta_{F|M,U}, \theta_{F|N,U}, \theta_N)$, where $\theta_{F|M,A} = P(F \mid X_C = M, X_S = A)$, $\theta_{F|N,A} = P(F \mid X_C = N, X_S = A)$, $\theta_{F|M,U} = P(F \mid X_C = M, X_S = U)$, $\theta_{F|N,U} = P(F \mid X_C =$

$N, X_S = U$), and $\theta_N = P(N)$ is the marginal probability of non-compliance, which under $G_3$is independent of seating type.

This separation of structure and strength in the Bayesian Network representation of theories is the central feature of the model, because it formalizes the two possible levels of interpretation of strategic information, corresponding to assessments about "what are the causal relationships that drive strategy?" and "what are the causal relationships that drive strategy like?" (Goldberg & Singell, 2024; Goldberg, 2011). This distinction is the basis for the different ways that decision-makers and employees form beliefs and disagree about strategies in their organization. Both the structure of the theory held by organizational actors G_k and the parameter space θ_k are informed by beliefs that organizational actors hold about the strategic environment, and the next section formalizes these beliefs and the process through which they are formed.

2.3 Two Levels of Belief: Theory Structure and Parameter Priors

With the theory defined as a causal graph structure actors G_k and their associated strength of causality parameter space θ_k, I now formalize the beliefs that actors hold about the strategic environment, where the belief state of actor i (B_i) is defined as the set of object pair vectors of how likely each potential causal graph G_k is, represented by the weight parameter $\omega_{i,k}$, and, for a given graph G_k , the probability distribution of different action-outcome combinations, represented by the collection of parameter priors $\mu_i(\theta_k|G_k)$.

$$B_i = (\{\omega_{i,k}\}_{G_k \epsilon G}, \{\, \mu_i(\theta_k|G_k)\}_{G_k \epsilon G})$$

The beliefs in B_i capture both which causal structures G_k actor i finds the most likely, and conditional on a graph G_k, the distribution of likely outcomes. In order to populate these beliefs, actors use their observational experience in the organization, and the next two subsections detail

specifically how an actor like the COO or the CCBO would use his observations to update both weights on a likely graph $\omega_{i,k}$ and the distribution of likely outcomes given actions $\mu_i(\theta_k|G_k)$.

2.3.1 Updating Beliefs About Theory Structures Based on Observations

The weights $\{\omega_{i,k}\}_{G_k \epsilon G}$ about which theory structures are likely are updated from observational evidence via Bayes' rule. When actor i observes the set or a subset of the variables $O_i \subseteq \{X_S, X_C, X_T\}$ their theory weights update as below for each theory G_k:

$$\omega'_{i,k} \propto \omega_{i,k} \cdot \int_{\theta_k} P(O_i|G_k, \theta_k)\, \mu_i(\theta_k|G_k) d\theta_k$$

The graph G_k that best explains what actor i has observed will have the greatest posterior weight $\omega'_{i,k}$, and will be the theory that the actor holds. All organizational actors will use their own observational experience that may or may not differ from each other, like the COO and the CCBO observing seating type and customer compliance differentially, to determine which theory structure is the most likely. An actor who assigns a high weight $\omega_{i,1}$ to G_1believes seating type directly causes turnaround time; an actor who assigns high weight $\omega_{i,2}$ to G_2believes compliance mediates this relationship. Two actors can agree on the full candidate set $\mathcal{G}$ yet disagree sharply in how they distribute their theory weights across it. If observation differs by employees, this rule is likely to push posterior theory weights $\{\omega'_{i,k}\}_{G_k \epsilon G}$ toward different candidate graphs.

2.3.2 Updating Beliefs About Parameter Priors Based on Observations

The parameter priors $\mu_i(\theta_k|G_k)$ about the distribution of which causal strength parameters θ_k make the observed data more likely given a data-generating graph G_k are also updated in accordance with Bayes' rule. When actor i observes the set or a subset of the variables $O_i \subseteq \{X_S, X_C, X_T\}$ their parameter priors update below for each graph G_k:

$$\mu'_i(\theta_k|G_k) \propto P(O_i|G_k, \theta_k) \cdot \mu_i(\theta_k|G_k)$$

This updating reflects the idea that conditional on the theory G_k being correct, the observations of an actor i (O_i) provide evidence about which causal strength parameters are likely, where θ_k parameter values that would have made the observed data more likely receive a greater posterior weight $\mu'_i(\theta_k|G_k)$.

All organizational actors update the likelihood of a given theory $\omega_{i,k}$, as well as their parameter priors $\mu_i(\theta_k|G_k)$, at the same time in response to new observations O_i, which is represented by the likelihood $P(O_i|G_k, \theta_k)$ entering both belief updating equations. This shared likelihood is what makes the two belief updates concurrent: a single observation simultaneously updates the actor's beliefs about which graph is correct (theory weight update, by marginalizing θ_k out) and about which parameter values are probable within each graph (parameter prior update, by conditioning on $G_k$and using the point likelihood in θ_k). Because beliefs about theories formalized as Bayesian networks contain these two concurrent objects, organizational actors who hold beliefs can disagree about them on two levels, which is the formalization I turn to next.

2.3.3 Formalizing Two Levels of Strategic Disagreement: Between and Within Theory

The beliefs about structure and strength of causal relationships that organizational actors who hold theories form allow for two distinct types of strategic disagreement in this model, between-theory and within-theory disagreement. I formalize a disagreement between actors i and j after observing their respective data along causal structure and strength as between-theory and within-theory disagreement.

Between-theory disagreement occurs when two actors' posterior theory weight collections differ, such that the posterior distribution for actor i and actor j are not equal, $\omega'_{i,k} \neq \omega'_{j,k}$ for some $G_k \in G$. Actors i and j do not share an agreed upon causal structure, because they

assign different probabilities to competing structural models. This represents situations where for example the COO and the CCBO's do not agree about how seating type ultimately causes turnaround time at the gate, since there is no consensus on whether customer compliance is an important intermediary variable in the underlying causal structure of the strategic environment.

Within-theory disagreement occurs when two actors i and j share the most likely theory structure, $\underset{k}{\operatorname{argmax}}\, \omega'_{i,k} = \underset{k}{\operatorname{argmax}}\, \omega'_{j,k} = k^*$, but their posterior parameter priors differ within that graph $\mu'_i(\theta_k|G_k) \neq \mu'_j(\theta_k|G_k)$ for the most likely theory G_k. This disagreement means that actors i and j agree about the cause structure of the strategic environment but disagree about the magnitude (including direction) of the effects within it.

With the way that actors update beliefs formalized into the two-level system of theory structure and strength, giving a clear two-level structure of disagreement, I now turn to how decision-makers select strategic actions, and how this process might generate between-theory disagreement for employees whose experience is divided.

2.4 Theory-Based Strategic Action: The CEO Selects a Strategic Action Using the Do-Operator

I now characterize the distinct role of the decision-maker in the strategic process, who holds a belief B_i about the strategic environment and uses it to select a strategic action, deciding which intervention to impose. Employees hold their own belief states B_i and use them to evaluate the CEO's proposed action against their own beliefs, which I will discuss further in the Section 2.5. However, this distinction between the CEO selecting strategic action and the employees evaluating is an important source of potential disagreement and is modeled in the decision-maker's selection of strategic action using the do-operator (see Ryall & Sorenson, 2025).

The do-operator $do(X_S = x)$, formalizes the way a decision-maker decides a strategic action, where the CEO for example sets the seating type to assigned seating by deliberate action, and severs all causal arrows into X_S which removes the influence of any upstream causes of the seating type variable that is being intervened on. This formalism quantifies a fundamental distinction in Pearl's (2009) causal framework, the difference between doing versus observing. The observational conditional $P(X_T \mid X_S = A)$ describes the distribution of turnaround time among cases where assigned seating is observed, which can conflate the causal effect of seating type with any other factors that independently influence both which seating policy is in use and how quickly turnarounds occur. The interventional distribution $P(X_T \mid do(X_S = A))$, by contrast, describes the distribution of turnaround time when seating type is set by deliberate action, which holds all other causal relationships fixed and isolates the effect of the strategic action itself.

The difference between doing versus seeing is foundational to the model because employees form their belief states from passive observation, reasoning about $P(X_T \mid X_S)$, while the CEO's strategic proposal is an interventional claim about $P(X_T \mid do(X_S))$. The gap between these two quantities is a source of systematic disagreement that the model makes precise. In this paper, because all potential graphs considered have seating type X_S as an exogenous factor with no causes in the consideration set that the CEO can freely set, no confounding paths exist between X_S and X_T, meaning observational and interventional distributions are equivalent.[1] However, I retain do-notation throughout to mark that the CEO's action is an intervention,

[1] $P(F|do(X_S = A) = P(F|X_S = A)$ under G_1, G_2, G_3. In a richer model allowing for confounding — for example, if scheduling pressure or route characteristics affected both which seating policy Southwest could feasibly implement and the baseline probability of fast turnaround — the backdoor criterion would not be satisfied, and $P(F|do(X_S = A)) \neq P(F|X_S = A)$ (see Ryall & Sorenson, 2025 for another example of satisfying the backdoor criterion). In that case the do-notation would alter the quantitative results, and the gap between doing and observing would generate an additional source of disagreement beyond those analyzed here. Future work should examine scenarios where exogeneity of the strategic action cannot be assumed.

because the conceptual distinction between doing and observing is important for strategic theories, and future work should consider how this difference can create meaningful divergence in strategic understandings.

In order for the decision-maker to select which strategic action he would like to do, he specifies two things: the graph G_k with the top weight $\omega_{i,k}$ and a constraint on the parameter priors $\mu_i(\theta_k|G_k)$ that makes a strategic action more likely to generate the strategic outcome of interest. In my model, because the focus of this paper is on employee disagreement conditional on a decision-maker holding a strategic theory, I assume which theory and strategic action the decision-maker takes.

Using the motivating example, I assume that the CEO holds graph $G_1: X_S \rightarrow X_T$, where seating drives turnaround directly, which means that his theory weight collection follows:

$$\omega_{CEO,1} > \omega_{CEO,2}, \omega_{CEO,3}$$

For his parameter priors, I assume that $\mu_{CEO}(\theta_1 \mid G_1)$ assigns high probability to parameter values where assigned seating raises the probability of fast turnaround, which is formalized as satisfying the relative probability constraint below:

$$P\big(F\big|do(X_S = A)\big) > P\big(F\big|do(X_S = U)\big)$$

Meaning that the CEO reasons that an intervention on X_S will shift the distribution of X_T in the desired direction. Thus, the CEO's proposed strategic action is:

$$do(X_S = A)$$

The CEO's theory, represented by both the directed acyclic graph (DAG) and parameter priors constrains the parameter space θ_1 to values that are consistent with the directional claim that assigned seating is more likely to lead to faster turnaround times than unassigned seating. I denote this restricted space as $\Theta_k \subset (0,1)^{|\theta_k|}$, which represents the set of parameter values

under the graph G_k consistent with the CEO's causal claim. I also formalize Θ_k's complement Θtilde_k, which is the null space of parameter values where the CEO's seating type action has no effect on turnaround time at the gate given graph G_k. While the CEO holds graph G_1 as the most likely, his theory constrains the parameter space differently contingent on which graph an employee finds the most credible, thus I formalize how his theory constrains the parameter space for each G_k with Θ_k and Θtilde_k.

$$\text{Under } G_1\text{: } \Theta_1 = \{\theta_1 : \theta_{FA} > \theta_{FU}\}\text{ , } \Theta\text{tilde}_1 = \{\theta_1 : \theta_{FA} = \theta_{FU}\}$$

Under the structure of G_1 the CEO's claim requires that assigned seating directly raises the unconditional probability of fast turnaround time at the gate.

$$\text{Under } G_2\text{: } \Theta_2 = \{\theta_2 : \theta_{N|A} < \theta_{N|U}\}\text{ , } \Theta\text{tilde}_2 = \{\theta_2 : \theta_{N|A} = \theta_{N|U}\}$$

Under the mediated structure of G_2, the CEO's claim that assigned seating will increase turnaround speed operates through the compliance channel: assigned seating must reduce non-compliance, which in turn improves turnaround. The restriction concentrates attention on the region where assigned seating produces lower non-compliance than unassigned seating.

$$\text{Under } G_3\text{: } \Theta_3 = \{\theta_3 : \theta_{F|M,A} > \theta_{F|M,U} \text{ or } \theta_{F|N,A} > \theta_{F|N,U}\}\text{ , } \Theta\text{tilde}_3 = \{\theta_3 : \theta_{F|M,A} = \theta_{F|M,U}\text{ , } \theta_{F|N,A} = \theta_{F|N,U}\}$$

Under the collider structure of G_3, the CEO's claim that assigned seating will increase turnaround speed requires that seating type has a direct positive effect on turnaround within at least one compliance state. The restriction concentrates attention on parameter values where at least one within-compliance contribution of seating type is positive.

Because the restrictions of the parameter space Θ_k and the definition of the null space Θtilde_k are both specific to the graph held G_k, the same theoretical claim generates different restrictions and interpretations under different graph structures. This means that an actor's

assessment of whether the CEO's claim is credible depends on which graph their theory weight finds most credible, and whether the parameter values they find probable under that graph fall within Θ_k or nearer its null complement Θtilde_k. The same observed outcome can confirm the null under one graph while supporting the theory under another, a point that becomes consequential when considering that under the division of experience in organizations, employees are likely to observe different parts of the strategic environment which may naturally lead to them holding different graphs. I turn to this division of labor and its role in theory-based strategic evaluation for employees next.

2.5 Theory-Based Strategic Evaluation: The Division of Labor and Situated Observation

I now turn to formalizing how employees use their belief B_i which identifies the structure and strength of their causal theory to evaluate the CEO's proposed action. Consistent with the attention-based view (Ocasio, 1997), I assume that employees' observations of their organization's strategy is limited by the division of labor. The COO and the CCBO receive the proposed action $do(X_S = A)$ and assess whether it is consistent with the causal understanding they have formed through passive observation. This evaluation process happens at the observational level, where they observe the seating type and its association with other variables through their organizational experience and update their belief state B_i using Bayes' rule as defined in section 2.3 to form a judgement about the CEO's proposed strategic action.

Under the division of labor, employees attend to different parts of the strategic state space. The COO, whose role is concentrated on operational processes, observes the relationship between seating policy and gate operations. The CCBO, whose role is concentrated on customer experience, observes the relationship between customer behavior and gate outcomes. Neither employee observes the full strategic state space $\{X_T, X_S, X_C\}$. Formally, each employee (i)

observes a subset $O_i \subsetneq \{X_T, X_S, X_C\}$ of the strategic state space, determined by their organizational role. This is represented in the Southwest example as:

$$O_{COO} = \{X_S, X_T\} \quad O_{CCBO} = \{X_C, X_T\}$$

The COO observes the seating type and turnaround time, but not customer compliance. The CCBO observe customer compliance and turnaround time, but not seating type. Crucially, both employees observe the strategic outcome of turnaround time at the gate X_T, but they observe different potential causal antecedents to that outcome.

Because each employee observes only a subset of the variables, the remaining variables are effectively latent from that employee's perspective. When evaluating candidate causal structures, each employee therefore compares the marginal likelihood of their observed data under each graph, integrating over both parameter uncertainty and the unobserved variables. Distinct causal graphs may produce identical marginal distributions over the observed variables once unobserved nodes are marginalized. When this occurs, the graphs are observationally equivalent from the perspective of that employee: no amount of observational data within the employee's role can distinguish between them.

This limited-observation structure is central to the model. The COO's observations marginalize over compliance (X_C), while the CCBO's observations marginalize over seating type (X_S). Because the candidate graphs differ primarily in the relationship between (X_S) and (X_C), neither employee directly observes the structural feature that distinguishes the graphs. As a result, the marginal likelihoods that drive belief updating differ systematically across employees, potentially leading them to favor different causal structures even when observing data generated by the same underlying environment.

This mechanism does not rely on actors observing different realizations of the same variables, but rather on observing different subsets of variables altogether. The division of labor therefore creates structurally different informational environments for employees, which in turn shapes which causal models are distinguishable from their perspective. I next show how this limited observation generates observational equivalence across candidate graphs and ultimately produces between-theory disagreement.

2.6 Calculating the Expected Value of a Strategic Outcome

I now defined the value function through which any actor assesses the expected strategic benefit of the proposed action $do(X_S = A)$. This value function is the formal object through which agreement or disagreement of beliefs B_i between actors manifests as a concrete difference in assessments of the effectiveness of the proposed strategy.

First, I define the expected probability of the desired strategic outcome of fast turnaround at the gate F for a single candidate graph G_k and parameter realization $\theta_k \in \Theta_k$ under the CEO's proposed intervention $do(X_S = A)$:

$$v_k(\theta_k) = P(F \mid do(X_S = A), G_k, \theta_k)$$

Since X_S is exogenous in all candidate graphs, $v_k(\theta_k) = P(F \mid X_S = A, G_k, \theta_k)$. The explicit form differs across graphs because the causal mechanism through which seating type affects turnaround differs structurally:

$$v_1(\theta_1) = \theta_{FA}$$

$$v_2(\theta_2) = \theta_{F|M}(1 - \theta_{N|A}) + \theta_{F|N} \cdot \theta_{N|A}$$

$$v_3(\theta_3) = \theta_{F|M,A}(1 - \theta_N) + \theta_{F|N,A} \cdot \theta_N$$

Under G_1, the expected probability of fast turnaround given assigned seating is θ_{FA} directly. Under G_2, it is a compliance-weighted average where the compliance distribution shifts

with the seating assignment through $\theta_{N|A}$. Under G_3, it is a compliance-weighted average where the compliance distribution θ_N is independent of seating type but the turnaround probability within each compliance state varies by seating type.

Because actor i is uncertain about the exact values of θ_k within each graph, they integrate $v_k(\theta_k)$ over their parameter prior $\mu_i(\theta_k \mid G_k)$, restricted to the space Θ_k consistent with the CEO's directional claim:

$$V_{\Theta_k,i} = \int_{\theta_k \in \Theta_k} v_k\,(\theta_k)\;\mu_i(\theta_k \mid G_k) d\theta_k$$

This corresponds to Camuffo, Gambardella, and Pignataro's (2024) V_Θ— the expected value of the proposed action under a given theory, integrating over the actor's parameter uncertainty within the theory-consistent region of the parameter space. An actor whose parameter prior concentrates near the null boundary of Θ_k will produce a lower $V_{\Theta_k,i}$ than one whose prior concentrates on large effect values, even if both are restricted to the same Θ_k.

Actor i's overall assessment of the CEO's proposed action integrates the within-graph expected values across all candidate graphs, weighted by the actor's posterior theory weight collection:

$$V_i = \sum_{G_k \in \mathcal{G}} \omega'_{i,k} \cdot V_{\Theta_k,i}$$

This is actor i's theory-weight-averaged expectation of the strategic outcome under $do(X_S = A)$. The theory weight collection $\{\omega'_{i,k}\}_{G_k \in \mathcal{G}}$ determines the credence assigned to each causal architecture; the parameter prior collection $\{\mu_i(\theta_k \mid G_k)\}_{G_k \in \mathcal{G}}$ determines the expected effect magnitude within each architecture. Strategic disagreement between actors $i$and $j$manifests as $V_i \neq V_j$. The results section below derives how differences in employee experience generates

such differences in V_i values, as well as analyzes how the CEO can design experiments to resolve them.

3. RESULTS

I show that limited and situated employee attention generated by the division of labor is sufficient to produce between-theory strategic disagreement. Because employees observe different subsets of the strategic state space, distinct causal graphs become observationally equivalent once unobserved variables are marginalized. When observational equivalence holds, posterior beliefs over causal structure are determined by priors. Combined with role-based priors, this produces systematic divergence in theory weights and therefore disagreement over the expected value of the CEO's proposed strategy.

3.1 Assumptions

The results that follow are derived from the model in section 2 under the following five assumptions about the CEO, COO, and CCBO and how their experience in the organization relates to the underlying strategic environment.

Assumption 1 (CEO's theory). The CEO holds graph $G_1: X_S \rightarrow X_T$ as most credible, with theory weights satisfying $\omega_{CEO,1} > \omega_{CEO,2}, \omega_{CEO,3}$, and his parameter prior $\mu_{CEO}(\theta_1 \mid G_1)$is concentrated on $\Theta_1 = \{\theta_1 : \theta_{FA} > \theta_{FU}\}$, reflecting his belief that assigned seating raises the probability of fast turnaround.

Assumption 2 (CEO's claim is correct). The CEO's interventional claim holds in the true strategic environment: $P(F \mid do(X_S = A)) > P(F \mid do(X_S = U))$, regardless of which candidate graph is true.

Assumption 3 (Role-based priors). Each actor assigns higher prior weight to candidate graphs whose node sets match their observation sets. Formally, since G_1 is the unique candidate graph

whose nodes $\{X_S, X_T\}$ lie entirely within $\mathcal{O}_{COO}$, the COO's prior satisfies $\omega_{COO,1} > \omega_{COO,2}, \omega_{COO,3}$. The CCBO's prior is unrestricted by this assumption, since all three candidate graphs include variables outside $\mathcal{O}_{CCBO} = \{X_C, X_T\}$.

Assumption 4 (Division of labor). The COO and CCBO observe subsets of the strategic state space as specified in Section 2.5: $\mathcal{O}_{COO} = \{X_S, X_T\}$ and $\mathcal{O}_{CBBO} = \{X_C, X_T\}$.

Assumption 5 (Causal relevance of X_C). The true causal graph contains a direct causal path $X_C \rightarrow X_T$— that is, the true graph is either G_2 or G_3. This means customer compliance genuinely affects turnaround time, so the CCBO's organizational experience yields a systematic association between X_C and X_T.

Assumptions 3–5 jointly characterize the informational asymmetry generated by the division of labor consistent with the attention-based view (Ocasio, 1997): the COO observes the seating–turnaround relationship, while the CCBO observes the compliance–turnaround relationship.

In combination with assumption 1, that the CEO holds a simple action to outcome theory ($G_1: X_S \rightarrow X_T$) assumption 5 outlines the cardinal error that the CEO makes when generating disagreement with his strategic action, mainly that he fails to consider and theorize about a strategic intermediary that is central to one of his core stakeholders (the CCBO) in determining the strategic outcome. This related to how the CEO resolves the disagreement between the COO and the CCBO, which I discuss next.

3.2 Two Levels of Disagreement and Their Resolution

Distinguishing between between-theory and within-theory disagreement is important because each type of disagreement requires different actions to resolve the disagreement. Particularly, within-theory disagreement, where actors share the same modal theory but hold

different parameter priors, is resolvable by standard experimentation. Because both actors assign highest credence to the same graph G_{k^*}, they interpret new experimental data through the same structural lens. The parameter prior updating rule $\mu_i'(\theta_k \mid G_k) \propto P(\mathcal{O}_i \mid G_k, \theta_k) \cdot \mu_i(\theta_k \mid G_k)$ applies symmetrically: both actors compute the same likelihood $P(\mathcal{O} \mid G_{k^*}, \theta_{k^*})$ for new observations, and sufficiently informative data will concentrate both posteriors $\mu_i'(\theta_{k^*} \mid G_{k^*})$ and $\mu_j'(\theta_{k^*} \mid G_{k^*})$ toward the same region of Θ_{k^*}. Within-theory disagreement therefore resolves naturally as evidence accumulates within the agreed causal structure.

However, between-theory disagreement is not resolvable this way, and this is the critical asymmetry. When two actors assign different theory weights to competing graphs, new data within any single graph's parameter space is interpreted conditionally on that graph being correct. The theory weight updating rule makes this precise: $\omega_{i,k}' \propto \omega_{i,k} \cdot \int_{\theta_k} P(\mathcal{O}_i \mid G_k, \theta_k)\mu_i(\theta_k \mid G_k)d\theta_k$. An actor who assigns low prior weight to G_1 will compute a low marginal likelihood for new observations under G_1 relative to their preferred graph, even if the data are consistent with G_1, because their parameter prior $\mu_i(\theta_1 \mid G_1)$ is diffuse and dilutes the likelihood. Additional evidence about $P(F \mid do(X_S = A))$, collected within the CEO's preferred graph G_1, does not shift the CCBO's theory weights toward G_1 if his prior already concentrates weight on G_2 and G_3. Between-theory disagreement therefore requires a qualitatively different resolution strategy: an experiment whose outcome is differentially probable under the competing candidate graphs, providing evidence about causal *structure* rather than effect *magnitude* within a given structure.

The paper's central claim is that limited and situated of employee attention, generated by the limited and divided attention of the division of labor, produces between-theory disagreement — not merely within-theory disagreement — as its default outcome. Because the COO and

CCBO observe subsets of the strategic state space, the observational likelihoods $P(\mathcal{O}_i \mid G_k, \theta_k)$ that drive theory weight updating emphasize different structural features of each candidate graph for each employee, pushing their posterior theory weight collections toward different G_k even from identical priors and the same true causal environment. Section 3.3 derives this result formally, and Section 3.4 analyzes the structurally diagnostic experiments that can resolve it.

3.3 The Sufficiency of Divided Labor for Strategic Disagreement

I show that the division of labor is sufficient to generate between-theory disagreement because it induces observational equivalence among competing causal graphs. When employees observe only subsets of the strategic state space, unobserved variables are marginalized, and distinct causal structures may generate identical marginal distributions over the observed variables. When this occurs, observational data cannot discriminate among the graphs, and posterior beliefs over theory structure are determined by priors.

Thus, disagreement arises not because employees interpret the same data differently, but because the data available to each employee renders different causal structures indistinguishable.

3.3.1 Observational Equivalence for the COO

The between-theory disagreement between the COO and the CCBO is generated by two formal properties of the model that must be established before deriving the employees' posterior theory weights: observational equivalence of graphs and the role of priors for each employee.

From the COO's observations $\mathcal{O}_{COO} = \{X_S, X_T\}$, graphs G_1, G_2, and G_3 are observationally equivalent, which means that priors determine the structure he finds the most compelling. Specifically, from the COO's perspective customer compliance (X_C) is unobserved and therefore marginalized. The observational likelihood is the marginal distribution $P(S, F)$

implied by each candidate graph, where the COO's observations integrate out compliance and calculate $P(F \mid S)$ for a given graph G_k and set of parameters θ_k.

- Under G_1: this is determined directly by calculating $P(F|S)$, because with parameter vector $\theta_1 = \{\theta_{F|S=A}, \theta_{F|S=U}\}$ the observable likelihood is $P(F|S) = \theta_{F|S}$
- Under G_2: marginalizing over compliance yields $P(F|S) = \sum_{c\in\{M,N\}} P(F|C\ )P(C|S)$, because with parameter vector $\theta_2 = \{\theta_{C|S=A}, \theta_{C|S=U}, \theta_{F|C=M}, \theta_{F|C=N}\}$ the observable likelihood is $P(F|S) = \theta_{F|C=M}\theta_{C=M|S} + \theta_{F|C=N}(1 - \theta_{C=M|S})$
- Under G_3: marginalizing over compliance yields $P(F|S) = \sum_{c\in\{M,N\}} P(F|C,S)P(C)$, because with parameter vector $\theta_3 = \{\theta_{F|S,C}, \theta_C\}$ the observable likelihood is $P(F|S) = \theta_{F|S,C=M}\theta_{C=M} + \theta_{F|S,C=N}\theta_{C=N}$

In all three cases, the resulting distribution over S and F can represent any conditional probability $P(F|S)$ through appropriate parameter choices θ_k. Consequently, all three candidate graphs generate the same observational distributions over the COO's observation set. From the COO's perspective G_1, G_2, and G_3 are observationally equivalent.

This means that observational data cannot discriminate among the graphs, and so the COO's posterior theory weights are determined entirely by the prior — the likelihood provides no updating force across graphs. By Assumption 3, that the COO's priors point him towards the graph that entirely contains his observational state space G_1: $X_S \rightarrow X_T$, his posteriors are:

$$\omega'_{COO,1} > \omega'_{COO,2}, \omega'_{COO,3}$$

The COO therefore selects G_1.

3.3.2 Observational Equivalence for the CCBO

However, from the CCBO's observations $\mathcal{O}_{CCBO} = \{X_C, X_T\}$, G_2 and G_3 are observationally equivalent while G_1 is not. This is because, from the CCBO's perspective, seating type (X_S) is

unobserved and thus marginalized, making the relevant likelihood the marginal distribution $P(C, F)$. Thus, the observational likelihood is the marginal distribution $P(C, F)$ implied by each candidate graph, where the CCBO's observations integrate out seating and calculate $P(F \mid C)$ for a given graph G_k and set of parameters θ_k.

- Under G_2, this is determined directly by calculating $P(F|C)$ because with parameter vector $\theta_2 = \{\theta_{C|S=A}, \theta_{C|S=U}, \theta_{F|C=M}, \theta_{F|C=N}\}$ the observable likelihood is $P(F|C) = \theta_{F|C}$
- Under G_3, marginalizing over seating type yields $P(F|C) = \sum_{S\in\{U,A\}} P(F|C,S)P(S)$, because with parameter vector $\theta_3 = \{\theta_{F|S,C}, \theta_S\}$ the observable likelihood is $P(F|S) = \theta_{F|S=A,C}\theta_{S=A} + \theta_{F|S=U,C}\theta_{S=U}$

Both graphs can generate any joint distribution over C and F through appropriate parameter choices θ_k. Formally, for any target distribution $(\phi_N, \phi_{F|N}, \phi_{F|M}) \in (0,1)^3$, there exist $\theta_2$and θ_3 such that $P(X_C, X_T \mid G_2, \theta_2) = P(X_C, X_T \mid G_3, \theta_3)$. To see this: under G_2, set $\theta_{N|A} = \theta_{N|U} = \phi_N$, $\theta_{F|N} = \phi_{F|N}$, $\theta_{F|M} = \phi_{F|M}$; under G_3, set $\theta_N = \phi_N$, $\theta_{F|N,A} = \theta_{F|N,U} = \phi_{F|N}$, $\theta_{F|M,A} = \theta_{F|M,U} = \phi_{F|M}$. Consequently, no data from $\mathcal{O}_{CBBO} = \{X_C, X_T\}$alone can discriminate between $G_2$and G_3, and from matched priors over the effective parameters, the CCBO's posterior theory weights for $G_2$and G_3 are equal. Thus, G_2 and G_3 are observationally equivalent. Under G_1 however, compliance plays no causal role, implying:

- Under G_1, marginalizing over seating type yields independence of customer compliance (C) and turnaround at gate (F) because with parameter vector $\theta_1 = \{\theta_{F|S=A}, \theta_{F|S=U}\}$ the observable likelihood is $P(F|C) = \sum_{S\in\{U,A\}} P(F|S)$, which does not depend on C.

This graph predicts independence between compliance and turnaround time. If the CCBO observes any systematic association between (X_C) and ($X_T = F$), G_1 is rejected. By assumption 5, which assumes that there is a true association between X_C and X_T, the CCBO will observe this systematic association. Thus, G_1 is observationally distinguishable from G_2 and G_3, while G_2 and G_3 remain observationally equivalent. This means that the posteriors of the CCBO will reject G_1 as the true graph and weight G_2 and G_3 higher as the potential true graphs, meaning:

$$\omega'_{CCBO,2}, \omega'_{CCBO,3} > \omega'_{CCBO,1}$$

Where $\omega'_{CBBO,1} \approx 0$. The CCBO thus selects either G_2 or G_3 as the true structure of causal relationships.

The division of labor therefore generates asymmetric observational equivalence. The COO cannot distinguish among any of the three graphs, but his priors lead him to believe G_1 is the correct graph, while the CCBO can eliminate G_1 but cannot distinguish between G_2 and G_3. Because each employee faces a different equivalence class of causal models, their posterior theory weights diverge even when observing data generated by the same environment.

3.3.3 Between Theory Disagreement

The divergence between the COO and the CCBO's theory structure arises purely from limited observation. Each employee's role renders different variables latent, and once those variables are marginalized, distinct causal structures collapse into indistinguishable observational predictions. As a result, priors and role-based experience determine which graph each employee ultimately favors, producing between-theory disagreement.

To formalize this disagreement, consider the weights for the COO and the CCBO produced above. The COO ends up with posterior weights that push him towards G_1. While the CCBO experience ends up pushing him away from G_1 as a viable graph, leading to $\omega'_{CCBO,1} \approx 0$. As a result of this, between-theory disagreement can be formalized between them as:

$$\underset{k}{\operatorname{argmax}}\, \omega'_{COO,k} = \omega'_{COO,1};\ \ \underset{k}{\operatorname{argmax}}\, \omega'_{CCBO,k} \neq \omega'_{CCBO,1};$$

$$\underset{k}{\operatorname{argmax}}\, \omega'_{COO,k} \neq \underset{k}{\operatorname{argmax}}\, \omega'_{CCBO,k}$$

This disagreement arises purely from limited observation combined with role-based priors. Both the COO and the CCBO observe the same causal reality, but their situated attention leads to disagreement on the true causal structure of the strategic, which ultimately leads them to evaluate the likely success of the CEO's proposed strategic action differently.

The divergent theory weight collections produce divergent overall expected values for the CEO's proposed action. The COO evaluates the proposed action primarily under G_1:

$$V_{COO} \approx \omega'_{COO,1} \cdot V_{\Theta_1,COO}$$

which is high when his parameter prior $\mu_{COO}(\theta_1 \mid G_1)$ concentrates within $\Theta_1 = \{\theta_1 : \theta_{FA} > \theta_{FU}\}$— reflecting direct experience of a seating-turnaround association. Since assumption 2 states that the CEO's strategic action of assigned seating increasing the speed of turnaround at the gate, the COO's experience will confirm the CEO's proposed strategic action, meaning that his evaluation will concentrate him on the positive relationship between the proposed action and outcome. Formally:

$$V_{COO}(A) \approx V_{\Theta_1,COO}(\mathrm{A}) \approx \int_{\Theta_1} P(F|S = A)\mu_{COO}(\theta_1 \mid G_1)\ d\theta_1$$

This calculation of the expected value reflects the fact that the COO only experiences parameter uncertainty, he does not have any graph uncertainty, since his role-related priors lead him to conclude G_1 is correct. Because assumption 2 states that there is a true relationship between assigned seating and faster turnaround, the COO's observations will confirm his graph and increase parameter weights for the relationship between assigned seating and faster turnaround, leading to a high valuation of the proposed action, and confirming the CEO's strategic action as effective.

The CCBO, on the other hand, evaluates the proposed action across G_2 and G_3:

$$V_{CCBO} \approx \omega'_{CCBO,2} \cdot V_{\Theta_2,CCBO} + \omega'_{CCBO,3} \cdot V_{\Theta_3,CCBO}$$

The CCBO's expected value is driven by whether his parameter priors $\mu_{CCBO}(\theta_2 \mid G_2)$ and $\mu_{CCBO}(\theta_3 \mid G_3)$ concentrate within Θ_2 and Θ_3 respectively — whether she believes the intervention will shift compliance (under G_2) or have a direct within-compliance effect (under G_3). Because the CCBO has no observational experience of X_S, his priors over the parameters that govern seating-compliance and seating-turnaround relationships are diffuse, placing substantial weight near $\widetilde{\Theta}_2$ and $\widetilde{\Theta}_3$. The CCBO's uncertainty in both structure and parameters push him towards the null hypothesis. This produces a lower and more uncertain V_{CBBO} than V_{COO}, constituting substantive disagreement about the expected strategic value of the CEO's proposed action — even though both employees began from identical priors and observed data generated by the same true causal environment. Formally:

$$V_{CCBO}(A) \approx \omega'_{CCBO,2} * \int_{\Theta_2} P(F|S = A)\mu_{CCBO}(\theta_2 \mid G_2)\, d\theta_2 + \omega'_{CCBO,3}$$

$$* \int_{\Theta_3} P(F|S = A)\mu_{CCBO}(\theta_3 \mid G_3)\, d\theta_3$$

where the nulls $\Theta\text{tilde}_2 = \{\theta_2 : \theta_{N|A} = \theta_{N|U}\}$ and $\Theta\text{tilde}_3 = \{\theta_3 : \theta_{F|M,A} = \theta_{F|M,U}, \theta_{F|N,A} = \theta_{F|N,U}\}$ look likely because of the joint uncertainty the CCBO experiences at the graph and parameter level due to his lack of observation of the relationship between seating type and turnaround speed. In not observing the seating type, the CCBO's priors over seating-related items are diffuse, and much prior mass lies near zero effect yielding:

$$V_{CCBO}(A) < V_{COO}(A)$$

Summarizing the above derivations, the division of labor as formalized in assumptions 3-5 produces:

1. Between-theory disagreement, where the COO selects G_1 and the CCBO selects either G_2 or G_3.
2. Which leads the expected values of whether the CEO's strategic action will generate the desired to strategic outcome to diverge between the COO and CCBO, where $V_{CCBO}(A) < V_{COO}(A)$.

Thus, the CEO's strategy generates between-theory disagreement despite being correct in expectation. This disagreement is structurally generated: both employees observe data from the same environment, but limited observation combined with role-based priors produces different causal beliefs and therefore different strategic evaluations. This means that the difference in the beliefs of the COO and the CCBO about the structure of the strategic environment will lead to

them having significantly different evaluations of the expected value of the CEO's proposed strategic action. Because this disagreement results from between-theory differences, the CEO will need to select an experiment that disambiguates the structure of the causal relationships between compliance and seating type in order to generate consensus on his chosen, and ultimately correct, strategic action, which I turn to next.

3.4 Selecting a Strategy to Resolve Strategic Disagreement

In order to resolve the disagreement of the COO and the CCBO, the CEO needs to fix the issue that led to it, which was, mainly, that his initial structure failed to theorize about the relationship between two potentially related causal antecedents and the desired outcome. To do this, he must design an experiment that directly tests the relationship between causal antecedents so that he can disambiguate between the different causal structures that are held by his employees.

In the example, the key distinguishing relationship to determine which structure is correct is between seating and compliance. The COO and CCBO face between-theory disagreement: their posterior theory weight collections concentrate on different candidate graphs, producing different overall expected values $V_{COO} \neq V_{CBBO}$. As established in Section 3.2, this disagreement cannot be resolved by implementing the CEO's proposed strategy $do(X_S = A)$and observing the turnaround outcome. Both the COO and CCBO will interpret whatever outcome is observed through the lens of the graph they find most credible: a favorable outcome confirms $G_1$for the COO, but the CCBO — weighting $G_2$and G_3— may attribute it to a favorable compliance state rather than to the seating change itself. An unfavorable outcome strengthens the CCBO's doubts but may be dismissed by the COO as sampling variation within G_1. Additional data on $P(F \mid$

$do(X_S = A))$ does not shift between-theory disagreement because it is interpreted conditionally on each actor's preferred graph.

What is required is a structurally diagnostic experiment: one whose outcome is differentially probable under the competing candidate graphs, providing evidence about causal *structure* rather than effect *magnitude* within a given structure. The key structural question distinguishing all three graphs is the relationship between $X_S$and X_C— whether seating type affects compliance. Under G_1, $X_C$is absent from the graph, predicting no relationship between seating type and compliance. Under G_2, seating causes compliance, predicting $P(N \mid do(X_S = A)) \neq P(N \mid do(X_S = U))$. Under G_3, seating and compliance are independent, predicting $P(N \mid do(X_S = A)) = P(N \mid do(X_S = U))$. The CEO should therefore test:

$$Does\ P(N \mid do(X_S = A)) \neq P(N \mid do(X_S = U))?$$

Since X_S is exogenous, $P(N \mid do(X_S = x)) = P(N \mid X_S = x)$, and the experiment reduces to measuring the compliance distribution under assigned versus unassigned seating in a controlled operational setting. This experiment is structurally diagnostic because its outcome provides evidence that discriminates among all three candidate graphs, shifting both employees' theory weight collections in the same direction and moving their overall expected values V_i toward alignment. I show what disambiguation looks like for both possible true underlying structures G_2 and G_3 below (recall under assumption 5 that G_1 is not a correct full specification of the strategic environment because there exists a true relationship between customer compliance and turnaround speed).

3.4.1 Disambiguation When Seating and Compliance Are Related

Suppose the experiment reveals that seating type and customer compliance are causally interdependent, specifically, that:

$$P(N \mid do(X_S = A)) < P(N \mid do(X_S = U))$$

This outcome allows for identify G_2 as the correct causal structure. This is because the outcome is inconsistent with G_1— which predicts no seating-compliance relationship — and with G_3— which predicts $X_S \perp X_C$. It is consistent only with $G_2: X_S \rightarrow X_C \rightarrow X_T$. Both employees' theory weights update strongly toward G_2:

$$\omega'_{i,2} \rightarrow 1 \text{for } i \in \{\text{COO, CCBO}\}$$

Under G_2, the CEO's causal claim constrains the parameter space to $\Theta_2 = \{\theta_2 : \theta_{N|A} < \theta_{N|U}\}$, and the experiment outcome directly informs both employees' parameter priors $\mu_i(\theta_2 \mid G_2)$by establishing that $\theta_{N|A} < \theta_{N|U}$with high posterior probability. Both employees' overall expected values converge:

$$V_{COO} \approx V_{CCBO} \approx V_{\Theta_2}$$

Agreement is generated at both the structure and parameter levels simultaneously. At the theory-level, both employees assign high weight to the same graph G_2, eliminating the structural disagreement. At the parameter-level, both employees update their parameter priors $\mu_i(\theta_2 \mid G_2)$from the same experimental outcome, moving their beliefs about the magnitude of the seating-compliance effect toward a shared posterior. As a secondary benefit, the CEO's own theory is corrected: his original causal claim was structured under $G_1: X_S \rightarrow X_T$, but the experiment reveals the operative mechanism is $G_2: X_S \rightarrow X_C \rightarrow X_T$. This refined causal

understanding identifies compliance as the mechanism through which the strategy operates, improving the CEO's ability to design implementation, monitor leading indicators, and anticipate failure modes.

3.4.2 Disambiguation When Seating and Compliance Are Independent

Suppose instead the experiment reveals:

$$P(N \mid do(X_S = A)) = P(N \mid do(X_S = U))$$

This outcome eliminates G_2, which requires a seating-compliance effect. Both employees' theory weights update away from G_2:

$$\omega'_{i,2} \to 0, \omega'_{i,1} + \omega'_{i,3} \to 1 \text{for } i \in \{\text{COO, CCBO}\}$$

In this case, the between-theory disagreement is improved, but not entirely resolved. Establishing that $P(N \mid do(X_S = A)) = P(N \mid do(X_S = U))$ eliminates the mediated interpretation G_2that the CCBO previously found credible alongside G_3. More importantly, this result forces the CCBO to evaluate the CEO's action under the same compliance distribution across seating conditions. Any difference in turnaround outcomes between assigned and unassigned seating cannot be attributed to compliance differences and must instead reflect a direct effect of seating type. The CCBO must therefore update his parameter priors under $G_3$to take seriously the possibility that $\theta_{F|M,A} > \theta_{F|M,U}$or $\theta_{F|N,A} > \theta_{F|N,U}$, shifting his priors toward Θ_3.

Note also that this outcome is consistent with Assumption 5 — compliance can affect turnaround (G_3 contains $X_C \to X_T$) without being affected by seating type. The CCBO's prior

experience of a compliance-turnaround association was not wrong; the experiment simply establishes that seating type does not operate through that channel.

In addition, within-theory disagreement may persist, however, this experiment generates alignment on trying the CEO's suggested strategy to resolve this disagreement. The COO's parameter prior $\mu_{COO}(\theta_1 \mid G_1)$ is shaped by direct observation of seating-turnaround associations, while the CCBO's parameter prior $\mu_{CCBO}(\theta_3 \mid G_3)$ over the seating-effect parameters is more diffuse for lack of direct observational experience of X_S. However, this residual within-theory disagreement is resolvable by standard experimentation: implementing the proposed strategy $do(X_S = A)$ and observing $P\big(F \mid do(X_S = A)\big)$versus $P(F \mid do(X_S = U))$ under controlled compliance conditions will update both employees' parameter priors toward a shared posterior. The CCBO will now accept this as a valid test of the CEO's theory because the structurally diagnostic experiment has eliminated the compliance-mediated interpretation she would otherwise have used to discount the result.

The two outcomes of the structurally diagnostic experiment therefore differ in their completeness of resolution. When seating and compliance are related, the experiment generates full agreement at both levels and corrects the CEO's understanding of the operative causal mechanism. When seating and compliance are independent, the experiment generates partial between-theory agreement — eliminating the primary source of between-theory disagreement — and converts any remaining disagreement into the within-theory form that standard experimentation can resolve. In both cases, the structurally diagnostic experiment targeting the X_S-X_C relationship is the correct first strategic move: not because it directly tests the CEO's proposed causal claim about turnaround time, but because it provides the graph-level evidence

necessary for employees with different observation sets to evaluate that claim from a shared causal framework.

4. DISCUSSION AND CONCLUSION

This paper shows that the division of labor is sufficient to generate fundamental strategic disagreement among employees who are causally theorizing about strategy, even when the proposed strategy is correct and employees reason rationally from their experience. The mechanism is the *observational equivalence problem*: when employees observe only a subset of the strategic environment, multiple competing theories about how the strategy works can produce identical predictions for what those employees will see. No amount of additional data within their observation set can help employees choose between theories that look the same from their vantage point. Because different employees who are assigned to different roles face different versions of this problem, they arrive at different beliefs about the causal architecture of the strategy, not because they interpret evidence differently, but because the evidence available to each of them is consistent with different sets of theories.

4.1 Theoretical Contributions

The paper makes three connected contributions. First, it extends the Theory-Based View (Felin & Zenger, 2017; Camuffo, Gambardella & Pignataro, 2024) from the benefits of theorizing for individual decision-makers to the coordination costs of theorizing in organizations. The TBV shows that holding explicit causal theories improves a decision-maker's ability to select effective strategies (Camuffo et al., 2020; Singell, 2026) This paper shows that the same explicitness creates a coordination problem: because theories make precise claims about which variables cause which outcomes, employees whose organizational roles give them access to

different parts of the causal environment will find different theories consistent with what they have observed. When strategy requires collective buy-in, the causal clarity that helps a decision-maker choose the right action can simultaneously prevent employees from agreeing that it is the right action.

Second, the paper provides a formal microfoundation for the coordination consequences of divided attention (Ocasio, 1997). The attention-based view establishes that organizational behavior is shaped by how attention is distributed. This paper makes precise what limited attention means for the formation of causal theories: employees with different observation sets face different equivalence classes of indistinguishable theories, and their structural beliefs are determined by the priors formed through their roles rather than by evidence. This formalizes a specific boundary condition for when divided attention becomes strategically costly — when the causal antecedents different employees attend to are genuinely related to the strategic outcome. When the decision-maker's theory is complete enough to encompass all causally relevant variables, divided attention produces only within-theory disagreement about effect sizes, which standard experimentation resolves. When the theory is incomplete, divided attention produces structural disagreement that experimentation cannot resolve without first establishing a shared causal framework.

Third, the paper identifies a structural source of implementation failure that conventional remedies do not address. Standard prescriptions for improving implementation — better communication, clearer goal-setting, incentive alignment — target information asymmetry and motivation, not the structural causal channel identified here (Carroll & Sørensen, 2021; Adner & Levinthal, 2024). A decision-maker who communicates the proposed action and expected outcome clearly but fails to establish a shared understanding of *how* that action generates the

outcome will face persistent structural disagreement among employees who have been differently exposed to the causal environment. Effective implementation of theory-based strategies therefore requires what the paper calls *structural alignment experiments* — experiments designed not to test whether the proposed action works, but to establish a shared causal framework within which the proposed action can be fairly evaluated.

4.2 Practical Implications

Three prescriptions follow directly from the results. First, decision-makers should diagnose the type of disagreement before selecting a resolution strategy. Within-theory disagreement — different beliefs about effect sizes within a shared causal architecture — is resolved by running the proposed experiment and observing the outcome. Between-theory disagreement — different beliefs about which variables cause which outcomes — is not resolved this way, because employees with different structural beliefs interpret identical experimental evidence through different causal lenses. The practical diagnostic is simple: ask employees what evidence would change their assessment. Within-theory disagreement produces convergent answers; between-theory disagreement produces divergent ones, because employees are asking different causal questions about the same proposed action.

Second, when structural disagreement is present, the first experiment to run is not the strategy itself but a test of the relationship between the causal antecedents that different employees attend to. In the Southwest example, the CEO should test whether changing seating type shifts customer compliance before testing whether it shifts gate speed. This structurally diagnostic experiment produces evidence that is differentially informative about causal architecture, giving employees from both vantage points a basis for jointly updating their structural beliefs. Whatever its outcome, it eliminates at least one candidate graph from

contention and moves the remaining disagreement into the within-theory form that standard experimentation can resolve.

Third, decision-makers should build theories that encompass the causal variables their key stakeholders observe. Theories that omit variables central to the experience of employees whose support is needed for implementation are more likely to generate structural disagreement, because those employees will find the decision-maker's preferred causal mechanism unrecognizable from their vantage point. Theory formulation in organizations should therefore be a consultative process — one that incorporates the causal variables key stakeholders attend to, even when those variables do not appear in the decision-maker's preferred causal path from action to outcome.

4.3 Limitations and Future Research

This model has several limitations worth highlighting. For example, the model assumes a fixed, shared set of candidate graphs — all actors reason over the same universe of causal theories. In practice, actors may not share awareness of all candidates, and extending the model to allow for heterogeneous candidate sets and varying unawareness (i.e. see Bryan, Ryall & Schipper, 2022) would likely reveal additional sources of structural disagreement. This model also treats observation sets as fixed by organizational role, whereas real observation sets overlap across roles and change over time through cross-functional interaction and information system design. In addition, as with some previous work, this model assumes the proposed strategic action is exogenous, the backdoor criterion is satisfied and observational and interventional distributions coincide (see Ryall & Sorenson, 2025). In settings where the proposed action is influenced by factors that also affect the outcome, an additional source of disagreement arises which could be a fruitful pathway for future research.

Finally, this paper generates testable predictions, which should be empirically explored. Organizations whose decision-makers hold structurally incomplete theories that omit causally relevant intermediaries that key stakeholders attend to should exhibit more persistent implementation failures, even controlling for the correctness of the proposed action. Structurally diagnostic experiments that test relationships between causal antecedents rather than the direct action-outcome relationship should generate better organizational alignment than direct strategy implementation when structural disagreement is present. Both predictions could be tested in settings where the causal structure of the strategic environment can be observed or controlled, and where measures of employee causal beliefs can be elicited before and after organizational experiments.

4.4 Conclusion

Organizations divide labor to enable execution, but in doing so they divide the causal experience of employees in ways that generate structural strategic disagreement. The division of labor makes different causal theories observationally indistinguishable from different employees' vantage points, and because employees face different equivalence classes of indistinguishable theories, their structural beliefs diverge even when they observe data from the same environment and reason appropriately. This disagreement cannot be resolved by implementing the proposed strategy and observing the outcome. It requires a structurally diagnostic experiment — one that tests the relationship between the causal antecedents different employees attend to, breaks the observational equivalence that sustains structural disagreement, and gives all employees a shared causal framework within which the proposed strategy can be fairly evaluated. Selecting the right strategic action is necessary for strategic success, but creating shared strategic understandings is what makes successful strategic implementation possible.

WORK CITED

Adner R, Levinthal DA (2024) Strategy Experiments in Nonexperimental Settings: Challenges of Theory, Inference, and Persuasion in Business Strategy. *Strategy Science*.

Anon Value Capture in the Face of Known and Unknown Unknowns. *Strategy Science*.

Camuffo A, Cordova A, Gambardella A, Spina C (2020) A Scientific Approach to Entrepreneurial Decision Making: Evidence from a Randomized Control Trial. *Management Science* 66(2):564–586.

Camuffo A, Gambardella A, Pignataro A (2024) Theory-Driven Strategic Management Decisions. *Strategy Science*.

Carroll GR, Sørensen JB (2021) *Making Great Strategy: Arguing for Organizational Advantage* (Columbia Business School Publishing, New York City).

Ehrig T, Schmidt J (2022) Theory-based learning and experimentation: How strategists can systematically generate knowledge at the edge between the known and the unknown. *Strategic Management Journal* 43(7):1287–1318.

Felin T, Gambardella A, Zenger T (2024) Theory-Based Decisions: Foundations and Introduction. *Strategy Science* 9(4):297–310.

Felin T, Zenger TR (2017) The Theory-Based View: Economic Actors as Theorists. *Strategy Science* 2(4):258–271.

Goldberg A (2011) Mapping Shared Understandings Using Relational Class Analysis: The Case of the Cultural Omnivore Reexamined. *American Journal of Sociology* 116(5):1397–1436.

Goldberg A, Singell MH (2024) The Sociology of Interpretation.

Joseph J, Gaba V (2020) Organizational Structure, Information Processing, and Decision-Making: A Retrospective and Road Map for Research. *ANNALS* 14(1):267–302.

Ocasio W (1997) Towards an Attention-Based View of the Firm. *Strategic Management Journal* 18:187–206.

Pearl J (2009) *Causality: Models, Reasoning and Inference* 2nd edition. (Cambridge University Press, Cambridge, U.K. ; New York).

Ryall MD, Sorenson O (2025) The Causal Inference Problem: When Can Managers Use Data to Inform Decisions and How Can Organization Design Help Them? *AMR*:amr.2022.0548.

Simon HA (1997) *Administrative Behavior, 4th Edition* (Free Press, New York).

Singell, Madison (2026) Any Old Theory Will Do: Why Cause-and-Effect Performance Links Form Parsimonious Mental Models of Complex Strategic Environments**.** *Preprint*.

Sorenson O (2024) Theory, Search, and Learning. *Strategy Science*.